5. ANALYSIS OF INVENTORIES

Inventory Measurement

	IFRS	GAAP
计量减值判断	The lower of cost or net realisable value	The lower of cost or market
	NRV=selling price - selling cost	market取值: 1.If replacement cost （RC）> NRV Market=NRV 2.If replacement cost（RC）< NRV - normal profit margin Market=NRV - normal profit margin 3.NRV - normal profit margin < RC < NRV Market=replacement cost
确认减值	If Cost > NRV •Inventory written down to the NRV on the B/S •COGS↑ （I/S）	If Cost > Market •Inventory written down to the market on the B/S •COGS↑ （I/S）
减值转回	可以转回，但是只能转回到原值	不可以转回

Agriculture product is an exception that allows using NRV even if > cost
There are some exceptions under US GAAP for inventory valued using the retail method, which would not apply to a manufacturing company or for inventory valued under LIFO.
if a company not using the Retail Inventory Method (RIM) under US GAAP, then US GAAP and IFRS follow the same rule, which is min(cost, NRV)

FIFO is an acceptable, but not preferred, method under IFRS, IFRS prefer Weighted Average Cost

- **FIFO assumes that the most recent purchases remain in inventory**:
  - Since FIFO sells the **oldest** inventory first, the remaining inventory consists of the most recently purchased items.
  - These recently purchased items are valued at **near-current market prices** (assuming stable or rising prices).

Inventory turnover = Cost of sales/historical Average inventory

Why Is There a LIFO Reserve but No FIFO Reserve?
The term **LIFO Reserve** exists because companies that use **LIFO (Last-In, First-Out)** must disclose the difference between their **LIFO inventory valuation and what it would have been under FIFO (First-In, First-Out)**. However, there is **no need for a "FIFO Reserve"** because FIFO itself is considered the standard benchmark for inventory valuation.

A **decline in demand** for a company's products suggests a **strong likelihood of future inventory write-downs** because it can negatively impact both **selling prices** and **inventory**

turnover

"cost" refers to COGS, the **historical cost of the inventory**, which includes the expenses incurred to acquire or produce the inventory

Any cost after finished goods are not COGS but selling cost

Selling Costs:

- These are costs directly related to selling the product, such as:
  - Shipping, storage or freight-out costs.
  - Sales commissions.
  - Marketing or promotional expenses.
  - Packaging costs specifically for sale (not production)

Retail method/ LIFO method (GAAP only):
Market = max(NRV−Normal Profit Margin, min(Replacement Cost ,NRV))

A write-down of inventory to NRV = average inventory decreases = turnover rate increase
= equity decreases = increase debt to equity ratio
= COGS increases = decrease gross margin, operating margin, and net margin

INFLATION IMPACT ON FIFO AND LIFO (cost aspect)

During Inflation:

Aspect	FIFO	LIFO
Net Income	Higher (older, lower costs matched with revenue)	Lower (newer, higher costs matched with revenue)
Inventory Value	Higher (newer, higher-cost inventory retained)	Lower (older, lower-cost inventory retained)
Tax Liability	Higher (due to higher net income)	Lower (due to lower net income)
COGS	Lower (uses older, cheaper inventory costs)	Higher (uses newer, more expensive inventory costs)

inventory value = unit * cost and is included in asset and equity

LIFO liquidation occurs when the company sells more inventory than it purchases, forcing the use of **older, lower-cost inventory layers**. This creates a temporary **profit boost** but also higher tax

The amount of loss reversals, from previously written-down inventory, recognized during the period are written in footnotes

Inventory Ratios

$$Inventory\,Turnover = \frac{COGS}{Average\ Inventory}$$

$$Days\ on\ hand\ = \frac{365}{Inventory\,Turnover}$$

Questions

Inventory measurement

Under which inventory cost flow assumption does inventory on the balance sheet best approximate its current cost?

ans:

FIFO. Under FIFO, ending inventory is made up of the most recent purchases, thereby providing a closer approximation of current cost.